

Honors Intro Micro | Homework 6

Due Thursday Dec. 9

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Send me your questions at taylorjweidman@pitt.edu. Answer clearly and completely; show your work so I can understand your thought process for partial credit; you are welcomed and encouraged to work in groups as long as your work is your own.

Question 1 | Wandmaker's Dilemma

Suppose the demand for wands is given by

$$P = 200 - Q$$

and there are two main wand sellers, Olivander and Gregorovitch. Olivander can make wands at a constant marginal cost of 20, while Gregorovitch can make wands at a constant marginal cost of 30. The marginal revenue for Olivander is

$$MR_O = 200 - 2q_O - q_G$$

and for Gregorovitch it is

$$MR_G = 200 - 2q_G - q_O$$

Units are in galleons and stones. Calculate the Nash equilibrium level of output for the two wandsellers and put it in the blanks below.

Note. *This Oligopoly is not symmetric. So the trick we used in class won't work here. Simply solve for both firm's optimization decisions separately.*

Part A. Olivander's Quantity

What is the Nash equilibrium quantity for Olivander?

Part B. Gregorovitch's Quantity

What is the Nash equilibrium quantity for Gregorovitch?

Part C. Equilibrium Price

What is the Nash equilibrium price in this market?

Part D. Subsidizing Wands

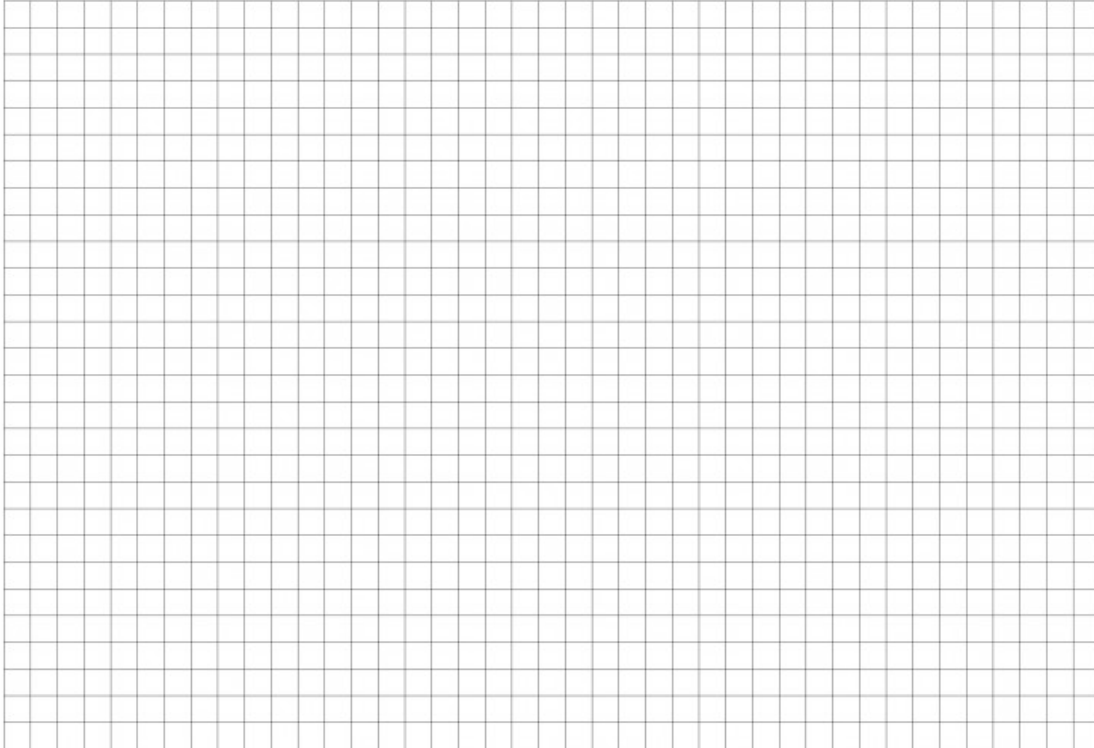
Many wizarding families had trouble affording a wand for their child when sending them to school. To make it easier for these families to send their children to wizarding school, the Ministry of Magic imposed a 10 galleon subsidy on wand sales. Model this as a 10 galleon decrease in the marginal cost of making a wand and find the Nash equilibrium quantity and price after the subsidy.

Question 2 | Luna's Problem

Luna divides her 100 galleons between pumkin pasties which cost 2 galleons and chocolate frogs which cost 4 galleons (both of which are normal goods). But surprisingly, an unusual fungus blight in Columbia causes the price of chocolate frogs to double right before the winter holiday season.

Part A

Draw Luna's budget constraint before the blight.



Part B

Show the effect of the blight on Luna's budget constraint on the graph from Part A.

Part C

Draw Luna's indifference curve on the graph from Part A and show the effect of the blight on Luna's optimal consumption bundle assuming that the substitution effect outweighs the income effect for pumpkin pasties.

Part D

In response to the price increase around the holiday season, the Ministry of Magic sent every wizard and witch a 10 galleon check in the mail to ensure families are able to afford the treat. Show the effect of the 10 galleons on Luna's optimal consumption bundle.

